

First Terminal Examination - 2082

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET - B

Pass Marks: 30

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Attempt all the questions.

Group 'A'

[11 × 1 = 11]

Rewrite the best alternative of each question in your answer sheet.

- The bi-phosphate of certain metal has formula MHPO_4 . Then the formula of metal chloride would be
 - MCl
 - MCl_3
 - MCl_2
 - M_2Cl_3
- Ethylene contains
 - 1σ and 5π bonds
 - 3σ and 3π bonds
 - 4σ and 2π bonds
 - 5σ and 1π bonds
- Which of the following molecules are polar?
 - H_2O
 - CH_3Cl
 - CHCl_3
 - CCl_4
 - i, ii and iii only
 - i, iii and iv only
 - i and ii only
 - i, iii and iv only
- The species having one unpaired electron is
 - CO
 - NO
 - CN^-
 - CO_2
- Which of the following correctly describes OF_2 ?
 - an acidic oxide
 - a basic oxide
 - an amphoteric oxide
 - not an oxide.
- What volume of CO_2 , measured at NTP, will be liberated when one gram of CaCO_3 is strongly heated?
 - 112ml
 - 224ml
 - 2.24 Litre
 - 1.12 Litre
- The value of 'R' depends upon
 - Nature of gas.
 - Unit of measurement
 - Mass of gas
 - Density of gas

8. Which of the following oxides is not an amphoteric oxide?
 a) PbO b) ZnO c) MgO d) Al₂O₃
9. Which of the following mixture of gases do not obey Dalton's law of partial pressure?
 a) O₂, CO₂ b) N₂, O₂ c) Cl₂, O₂ d) NH₃, HCl
10. Which of the following molecules is electron-deficient?
 a) PCl₃ b) PCl₅ c) BCl₃ d) NH₃
11. Higher number of oxygens is present in
 a) Neutral oxide b) Peroxide c) Superoxide d) Suboxide

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. Different laws of gases explain the macroscopic properties of gases, such as volume, pressure, and temperature etc.
 - a. State and explain the Boyle's law and write its one application. [3]
 - b. Calculate the pressure required to compress 2000 mL of a gas at 700 mm pressure and 20°C into a container of 100 mL capacity at temperature of 25°C. [2]
13. Nitrogen is the most important constituent of protein, hence all plants and animals contain nitrogen. Ammonia is the hydride of nitrogen and have various applications.
 - a. How would you explain the inertness of nitrogen? [1]
 - b. How is ammonia manufactured by the Haber process? Write the principle with reaction and explain the necessary conditions required for obtaining maximum yield of ammonia. [3]
 - c. Why can't we use conc. H₂SO₄ for drying of ammonia gas? [1]
14.
 - a. Define oxidation number. [1]
 - b. Calculate the oxidation number of underlined elements in the given compound [2]
 - i) KMnO₄ ii) SO₄²⁻
 - iii) LiAlH₄, iv) [Cu(NH₃)₄]₂SO₄
 - c. Identify the oxidizing and the reducing agent in the given reaction by indicating oxidation half and reduction half reaction. [2]

$$2\text{FeCl}_3 + \text{SnCl}_2 \rightarrow \text{SnCl}_4 + 2\text{FeCl}_2$$

15. Mole of substance can be express in terms of mass, volume at NTP and number of particles present in substance.

a. Complete the following table [3]

Moles of substance	In terms of mass	In terms of volume at NTP	In terms of number of particles
2 moles of CO ₂	-----	-----	-----
-----	80 gram of CH ₄	-----	-----

- b. Determine the percentage composition of hydrogen, oxygen and sulphur in H₂SO₄. [2]
16. Ozone is a highly reactive gas composed of three oxygen atoms. It naturally occurs in the stratosphere.
- a. How does chlorofluorocarbon deplete the ozone layer? Explain with chemical reactions. [2]
- b. Write the resonance structure of ozone. [1]
- c. What is meant by the tailing of mercury? Explain. [1]
- d. What type of oxide is Na₂O₂? Justify your answer. [1]
17. Electronegativity difference decides whether a covalent bond is polar or non-polar.
- a. CO₂ is linear, but H₂O is not why? Explain on the basis of dipole moment [2]
- b. Draw the Lewis electron dot structure of; [3]
- i) H₃PO₄ ii) H₂O and H₂O₂ iii) CO₃²⁻ and NH₄⁺
18. Valence shell electron pair repulsion (VSEPR) theory predicts molecular geometry based on electron pair repulsion.
- a. Using VSEPR theory, explain the shapes of BeF₂ and BF₃ molecules. [2]
- b. Why is the H–O–H bond angle in a water molecule comparatively larger than the H–S–H bond angle in a hydrogen sulphide molecule? [2]
- b. Mention one limitation of VSEPR theory. [1]
19. Combined gas equation is $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$
- a. Derive the equation. [3]
- b. A spherical balloon of 21 cm diameter is to be filled with hydrogen gas at NTP from a cylinder containing hydrogen gas at 20 atm, and 27 °C. If the cylinder can hold 2.82 litre of water, calculate the number of balloons that can be filled up. [2]

Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. Chemical bonding is the fundamental interaction that holds atom together to form molecules and extended structure. It arises from the redistribution of electrons between atoms and can be broadly classified into ionic, covalent, co-ordinate, metallic etc.
- a. Define valence electron and core electron. [1]
 - b. What is ionic bond? What are the conditions of formation of ionic bond? [2]
 - c. Write any two differences between σ and π bond. [2]
 - d. What is meant by non-polar covalent bond? How polarity is originated in polar covalent compound. [2]
 - e. What is dative bond? How is it formed? [1]
21. a. What are the basic postulates of Kinetic theory of gas? [4]
- b. Why do gases deviate from its ideal behaviour at low temperature and high pressure? [1]
- c. A cylinder of a gas contains 30 litres of gas under a pressure of 50 atm. Calculate the number of jars each of 400cc capacity which would be filled up from this cylinder at 750 mm of Hg pressure. [3]
22. Hydrogen gas is a direct source of clean energy and has applications in food industries, petrochemical refinery, cutting/welding, metallurgy, etc.
- a. What happens when a piece of zinc is added to potassium permanganate solution acidified with dil. H_2SO_4 ? [2]
 - b. Write the differences between nascent hydrogen and atomic hydrogen. [2]
 - c. Write one medical and one industrial application of hydrogen peroxide. [2]
 - d. Why is hydrogen fuel regarded as zero-emission fuel? [1]
 - e. What is heavy water? Write its one use. [1]

❧ The End ❧